the equation of the parabola w.r.t. XOY system is $Y^2 = 4aX$.

By shifting the origin $X = x - h$ and $Y = y - k$, the equation of the parabola with respect to old xoy system is

$$(y-k)^2 = 4a(x-h)$$

This is the general form of the standard equation of the parabola, which is open rightward.

Similarly the other general forms are

$(y-k)^2 = -4a(x-h)$ (open leftwards)

$(x-h)^2 = 4a(y-k)$ (open upwards)

$(x-h)^2 = -4a(y-k)$ (open downwards)

Example 3.31

Find the equation of the parabola whose focus is (1,3) and whose directrix is $x - y + 2 = 0$.

Solution

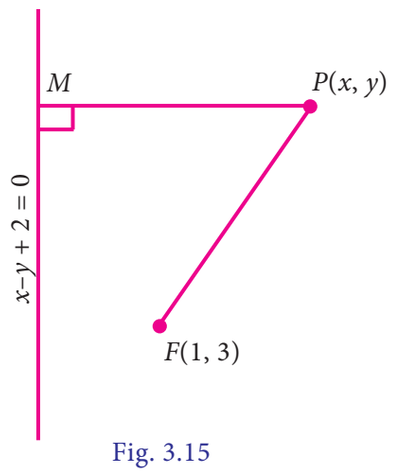


Fig. 3.15

F is (1, 3) and directrix is $x - y + 2 = 0$

$x - y + 2 = 0$

Let $P(x,y)$ be any point on the parabola.

For parabola $\frac{FP}{PM} = 1$

$FP = PM$

$$FP^2 = (x-1)^2 + (y-3)^2$$

$$= x^2 - 2x + 1 + y^2 - 6y + 9$$

$$= x^2 + y^2 - 2x - 6y + 10$$

$$PM = \pm\frac{(x-y+2)}{\sqrt{1+1}}$$

$$= \pm\frac{(x-y+2)}{\sqrt{2}}$$

$$PM^2 = \frac{(x-y+2)^2}{2}$$

$$= \frac{x^2+y^2+4-2xy-4y+4x}{2}$$

$$FP^2 = PM^2$$

$$2x^2 + 2y^2 - 4x - 12y + 20 = x^2 + y^2 - 2xy - 4y + 4x + 4$$

The required equation of the parabola is

$$\Rightarrow x^2 + y^2 + 2xy - 8x - 8y + 16 = 0$$

Example 3.32

Find the focus, the vertex, the equation of the directrix, the axis and the length of the latus rectum of the parabola $y^2 = -12x$

Solution

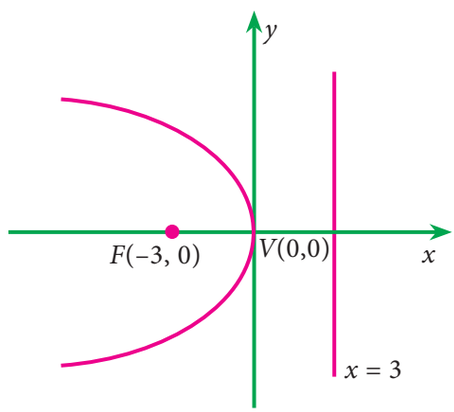


Fig. 3.16

The given equation is of the form $y^2 = -4ax$

where $4a = 12$, $a = 3$.

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